

Efficient Clothing Recognition and Visual Search Using Deep Embedding Spaces

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Abstract—In recent times, fashion recognition systems have gained popularity in the field of e-commerce, where these systems help in automatic classification of clothing items, and also support personalized recommendations for users. However, many existing systems depend on the quality images, and some of them also use metadata information, which may make them less effective in real shopping environments. In real life, images can be blurry, noisy, or have cluttered backgrounds. In this study, we propose an image-only pipeline for fashion recognition and recommendation, where EfficientNet V2, fine-tuned on a dataset of about 7,500 clothing images consisting of 15 unique categories, is used as the backbone model. The pipeline was evaluated on clean images as well as synthetically distorted images. Several preprocessing steps were applied during training, data augmentation techniques were also used, and label smoothing was added to improve generalization. The EfficientNet V2 model achieved 80.9% validation accuracy on clean images. When Gaussian blur and brightness jitter were applied, the accuracy remained 73.8%, which shows that the model performs reasonably well even with noisy inputs. In addition, deep feature embeddings were extracted from the second last layer of the network, and were used to build a visual recommendation system. The system achieved 90% Precision@1 when retrieving the most visually similar item. It achieved 80% Precision@5 for broader recommendation queries. These results show that image-only models can work well for practical applications, which may reduce the need for metadata information, while maintaining accuracy and robustness. Overall, the proposed framework provides a scalable and noise-tolerant solution, which supports intelligent fashion retail platforms and smart shopping applications.

Index Terms—Clothing classification, EfficientNet V2, Noise robustness, Deep embeddings, Visual recommendation, Pattern

recognition, E-commerce.

I. INTRODUCTION

Fashion recognition has become an integral application of several modern e-commerce systems, where it supports several applications, including automatic product cataloging, visual search, and personalized shopping recommendations. Its relevance and application for common users becomes much more significant, when common users can upload images directly to the system instead of being uploaded by professional photographers and then evaluated. However, it should be noted and observed that this brings a new challenge altogether as these user-generated images often differ significantly from clean product photos from a single source in real-world scenarios. The user generated and uploaded pictures may contain cluttered backgrounds, partial occlusions, varying camera angles, and sensor-related noise, among various other differences, where these factors are known to typically negatively affect classification performance and lead to noticeable drops in accuracy. Recent advances in computer vision and natural language processing indicate that modern deep learning backbones can still achieve competitive, better, and improved results despite these challenging and noisy conditions. However, this typically requires appropriate data augmentation strategies, data preprocessing techniques, extensive model training, and carefully designed evaluation procedures. Some studies have also incorporated detection-based pipelines and transfer learning techniques to

enhance the robustness of the model against these variations in the input images [1], [2].

At the same time, it was observed that the practical deployment in e-commerce platforms demands lightweight as well as computationally efficient models that can work in real-time and get results quickly. Although large architectures may offer strong accuracy, they often introduce high inference costs and take a lot of time to infer results from images input by users. Consequently, when studying recent research, we noticed that the modern literature has focused on compact architectures that aim to balance predictive performance and computational efficiency at the same time [3], [4]. In particular, several studies based on related image classification tasks showed that the convolutional neural network (CNN)-based transfer learning has demonstrated strong generalization capabilities, particularly when the training data is limited or is strongly visually diverse. For transfer learning based systems, fine-tuning pretrained models helps in showing better performance due to its ability to adapt to domain shifts in visually similar problems and therefore, improves the model robustness across varying data distributions as observed in various recent studies [5], [6].

Another important consideration is reliability under distorted or imperfect input conditions, which is a critical factor to consider under user input images as is the target input in this study. Several studies done on similar topics and input data, therefore, have evaluated model performance not only on clean preprocessed datasets, but have also evaluated under noisy or degraded image settings [7], [8]. Building upon these insights and target goals in mind, our study proposes a fully image-only pipeline for clothing recognition and recommendation. The proposed system adopts EfficientNetV2 as its backbone architecture and emphasizes robustness against blur and brightness distortions while maintaining computational efficiency.

Top-1 category prediction is useful for organizing products in online stores, but it has several known caveats that should be carefully considered. It is well-known that modern retail systems also need recommendation capabilities for attracting customers and personalizing to their needs. The typical example of one such feature providing this facility to the end user is that the users expect to see visually similar items through features like “you may also like”. One way to achieve this is through embedding-based retrieval, where the main approach is that the model learns feature representations where visually similar products appear close to each other. This approach allows the system to retrieve similar items without relying on additional metadata, and is known to work very well with such systems. Similar ideas have particularly been explored in personalized recommendation systems, where learning-based policies and ranking models are known to help tailor suggestions according to user preferences and are based on existing data [9], [10]. For real applications, it is well known that the accuracy alone is not enough and developers also want to train machine learning models and recommendation systems that are easier to interpret and analyze. Explainable AI methods are, therefore, becoming more common in machine learning systems and are commonly applied to various deep

learning and artificial intelligence systems [11]. Robustness is another important factor that is typically considered for such systems. The robustness condition measures how the various recommendation systems and machine learning models work consistently across different environments, input, and image conditions. One of the well-known and key observation in system robustness is that the bias and sensitivity to nuisance factors can affect system reliability [12]. Finally, the last important factor for practical deployment is the requirement of computational efficiency for deployment in real-time systems. The users can not spend a long time waiting for decisions to be inferred by the machine learning model, and therefore, systems must maintain good performance while operating on large product catalogs or limited hardware resources [13]. Typically, the user hardware capabilities are not built for accuracy but more for quick real-time inference with limited hardware resources, and therefore, can not handle extremely large models typically built for higher accuracy in modern day deep learning architectures. Based on these requirements, our framework combines EfficientNetV2 classification with deep feature embeddings for visual recommendation. The system is evaluated under synthetic distortions to better simulate real shopping scenarios.

In this work, we present an end-to-end image-only pipeline for fashion recognition and recommendation. The system is designed to work directly on RGB images without using any additional metadata.

- The EfficientNet V2 model is fine-tuned on a real-world RGB fashion dataset. The dataset contains 15 clothing categories and about 7.5K images. The model achieved 80.9% validation accuracy on clean images.
- We also evaluate the model under synthetic distortions such as Gaussian blur and brightness jitter. Under these conditions, the accuracy reduces to 73.8%.
- Deep feature embeddings are extracted from the network to build a visual recommendation system. The system can retrieve top-5 visually similar garments for a given query image.

The main contributions of this paper are summarized as follows:

- 1) Showing that EfficientNet V2 can provide strong baseline performance on real-world fashion images.
- 2) Studying how the classification accuracy changes under realistic noise conditions.
- 3) Developing a visual embedding-based recommendation system that does not depend on any metadata.

Recent research shows that large-scale data analytics and machine learning methods are becoming very important for building predictive and recognition systems in many application areas [14]. In computer vision tasks, optimizing pretrained convolutional neural networks has shown good improvements in classification performance while keeping the computational cost manageable [15]. Another important direction is explainable artificial intelligence. These methods help in making model decisions more transparent and easier to understand

in complex data-driven systems [16]–[18]. Because of these developments, there is increasing interest in deep learning pipelines that provide good accuracy while also being interpretable and practical for deployment.

A. Research Questions

To address these gaps, we investigate:

- 1) **RQ1:** What validation accuracy can a fine-tuned EfficientNet V2 achieve on a real-world RGB fashion dataset?
- 2) **RQ2:** How does synthetic noise (Gaussian blur + brightness jitter) affect classification performance?
- 3) **RQ3:** How effective are the feature embeddings from EfficientNet V2 for retrieving visually similar garments without metadata?

II. LITERATURE REVIEW

Previous studies in applied image-based and/or video-based recognition tasks indicate that transfer learning with modern CNN architectures provides a strong and practical baseline for multi-class classification problems. We know from modern literature that this hypothesis is especially true when the models are supported by suitable data augmentation, detailed data preprocessing, and careful fine-tuning strategies deployed in modern times [5], [19]. In vision-based applications, object detection and representation learning pipelines have demonstrated that robustness to acquisition noise and complex scene backgrounds can be achieved when appropriate backbone networks and training protocols are selected [1], [2]. At the same time, broader system-level decisions have included preprocessing steps, model size, computational efficiency, and evaluation under distribution shift [4], [7], [8]. For intelligent systems that directly interact with users, interpretability and accountability are also important considerations. This has encouraged the adoption of explainability techniques to better understand model decisions and to detect possible spurious correlations [11]. Related AI applications further show that integrating machine learning components into practical user workflows, such as assistants and interactive systems, requires models that are robust, user-friendly, and stable during inference [20], [21]. Moreover, segmentation approaches, including scene parsing and semantic segmentation, especially which are focused on pixel level identification and classification, can help isolate garment regions from cluttered backgrounds, which can then help improve feature quality as well as enhance embedding-based retrieval performance [22]. As a final point, the research in multimedia and deep feature modeling has highlighted that architectural design choices can significantly influence representation quality and downstream retrieval outcomes [23]–[25].

III. METHODOLOGY

A. Dataset

We use the “Clothes Dataset” from Kaggle¹, which contains 7 500 RGB images of 15 apparel categories collected “in the

wild”. Each image varies in resolution, pose, lighting, and background, making it ideal for robustness testing.

TABLE I
DATASET SPLIT AND IMAGE COUNTS

Total images	7 500
Categories	15
Training (80%)	6 000
Validation (20%)	1 500
Input size (after prep.)	224×224 px

Images are first resized to 256×256 px, then randomly cropped to 224×224 pixels during training; center-cropped at validation time. Then, we applied random horizontal flips, color jitter (brightness/contrast/saturation), and random erasing to increase robustness to variations in the images. Then, the pixel values are normalized to mean = [0.485,0.456,0.406], std = [0.229,0.224,0.225], matching ImageNet statistics.

B. Overall Workflow

Our end-to-end pipeline consists of five main stages. In the first stage, we performed Data Loading & Preprocessing, where we loaded images with the transforms. Then model training is performed, where we fine-tuned EfficientNet V2-S on the training split. Then validation is done, where we evaluated classification accuracy, precision, recall, and F1 on the cleaned validation set. This is followed by testing on noise robustness, where we applied Gaussian blur and brightness jitter to validation images and record accuracy drop. Finally, we performed embedding extraction and recommendation, where we extract 280 1-D feature vectors, and then retrieved top-K visually similar items via cosine similarity.

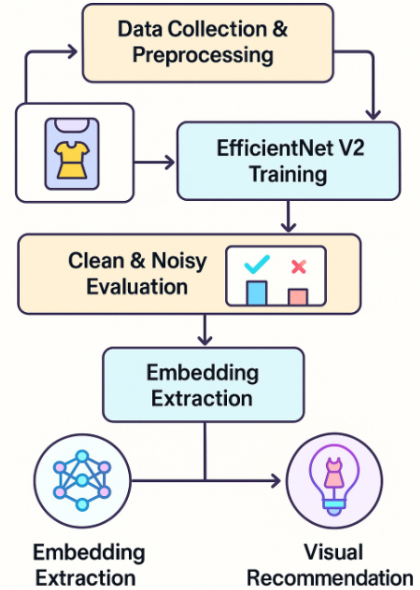


Fig. 1. Stylized workflow: (1) Preprocessing, (2) EfficientNet V2 training, (3) clean and noisy evaluation, (4) embedding extraction, (5) visual recommendation.

¹<https://www.kaggle.com/datasets/ryanbadai/clothes-dataset>

C. Model Architecture & Training

EfficientNet V2-S Classifier. We use the EfficientNet V2-S backbone, initialized with ImageNet weights. We replace the final classification head with:

- **Dropout(0.3):** Prevents overfitting by randomly zeroing 30% of features.
- **Linear layer:** Maps the 1 280-dim embedding to 15 class logits.
- **Softmax:** Converts logits to class probabilities.

Embedding Extraction. The penultimate 1 280-dimensional feature vector is used for retrieval. For a query image, we compute cosine similarity between its vector and all training-set vectors, then return the top-K most similar items.

TABLE II
TRAINING CONFIGURATION

Parameter	Value
Batch size	32
Epochs	15
Optimizer	Adam
Learning rate	1e-4
Weight decay	1e-5
Scheduler	CosineAnnealingLR
Label smoothing (CrossEntropy)	0.1
Device	2× T4 GPU

1) *Training Hyperparameters: Training Loop.* Each epoch consists of:

- 1) Forward pass on training batches, compute cross-entropy loss with label smoothing.
- 2) Backward pass and optimizer step.
- 3) Scheduler update per epoch.
- 4) Validation pass to record loss and accuracy.

Evaluation Metrics. We report Accuracy, which is the overall proportion of correctly classified images, as well as precision, recall, and F1-score, which were computed per class and averaged. Similarly, we measured the robustness drop, which is measured as the difference in accuracy between clean and noisy validation. We also used retrieval precision@K, which is qualitatively demonstrated with top-5 examples.

IV. RESULTS

We now present the empirical findings for each research question.

A. Classification Performance

We fine-tuned EfficientNet V2-S on our in-the-wild 15-class clothes dataset. After 15 epochs of training (see Fig. 3), the model converged to a clean-validation accuracy of 80.9%. We noted that the training loss steadily decreased and plateaued around 0.58, but the validation loss got stabilized near 1.10, which indicates good fit without overfitting.

Table III breaks down per-class precision, recall, and F1. We see that high-support classes such as “Jeans” (F1 = 0.911) and “Kaos” (F1 = 0.888) achieved particularly strong performance.

Fig. 4 visualizes these misclassifications: most errors occur between stylistically similar garments (e.g., “Blazer”

TABLE III
PER-CLASS PRECISION, RECALL, AND F1-SCORE ON CLEAN VALIDATION SET (1 500 IMAGES)

Class	Prec.	Rec.	F1	Supp.
Blazer	0.714	0.660	0.686	106
Celana_Panjang	0.856	0.865	0.860	96
Celana_Pendek	0.941	0.930	0.936	86
Gaun	0.752	0.738	0.745	103
Hoodie	0.874	0.874	0.874	103
Jaket	0.771	0.694	0.730	121
Jaket_Denim	0.827	0.966	0.891	89
Jaket_Olahraga	0.764	0.698	0.730	116
Jeans	0.915	0.907	0.911	107
Kaos	0.838	0.943	0.888	88
Kemeja	0.702	0.784	0.741	102
Mantel	0.763	0.724	0.743	98
Polo	0.767	0.690	0.726	100
Rok	0.788	0.830	0.808	94
Sweter	0.885	0.934	0.909	91
Avg./Total	0.807	0.809	0.807	1500

vs. “Jaket”), while classes with distinctive silhouettes (e.g., “Celana_Pendek”) cluster tightly on the diagonal. Overall, these results confirm that EfficientNet V2 provides a strong baseline for real-world clothing recognition, with room for targeted data augmentation or fine-tuning to disambiguate the closest categories.

B. Noise Robustness

Applying Gaussian blur and brightness jitter leads to an accuracy drop from 80.9% to 73.8%. Table IV and Figure 5 summarize these results.

TABLE IV
ACCURACY OF EFFICIENTNET V2 UNDER SYNTHETIC DISTORTIONS

Condition	Accuracy (%)
Clean images	80.9
Gaussian blur	73.8
Brightness jitter	74.0
Random background distortion	71.3

C. Embedding-Based Recommendation

To evaluate our pure-visual recommender, we extract 1 280-D feature vectors from EfficientNet V2 and compute cosine similarity to retrieve the top-K nearest neighbours for each query. Table V shows that even at $K = 1$, we achieve 90% precision, indicating that the single most similar item is correct nine times out of ten. As K increases, precision gracefully degrades to 80 % at $K = 5$, demonstrating a robust retrieval horizon.

V. CONCLUSION

In this work, we developed a fully image-only fashion recognition and recommendation pipeline based on EfficientNet V2. Our fine-tuned model achieved an overall validation

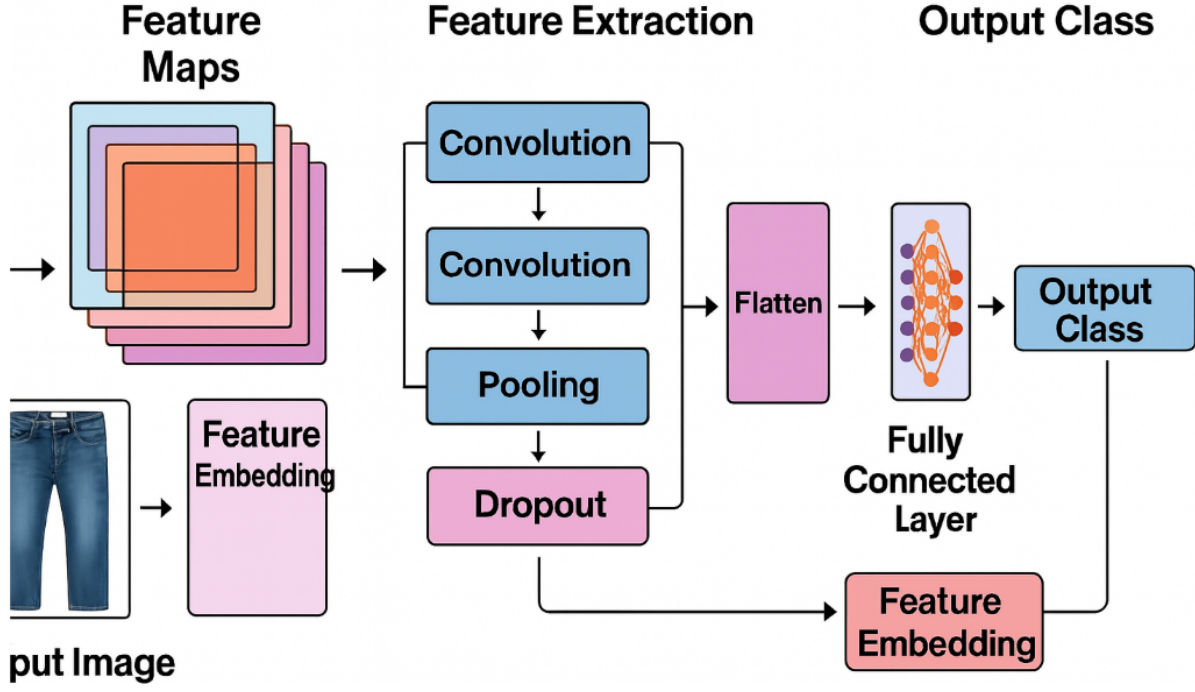


Fig. 2. EfficientNet V2-S classifier architecture, showing (a) input image, (b) backbone + dropout/head, (c) 1280-D embedding tap, and (d) example output thumbnails for “Jeans,” “Dress,” “Jacket.”

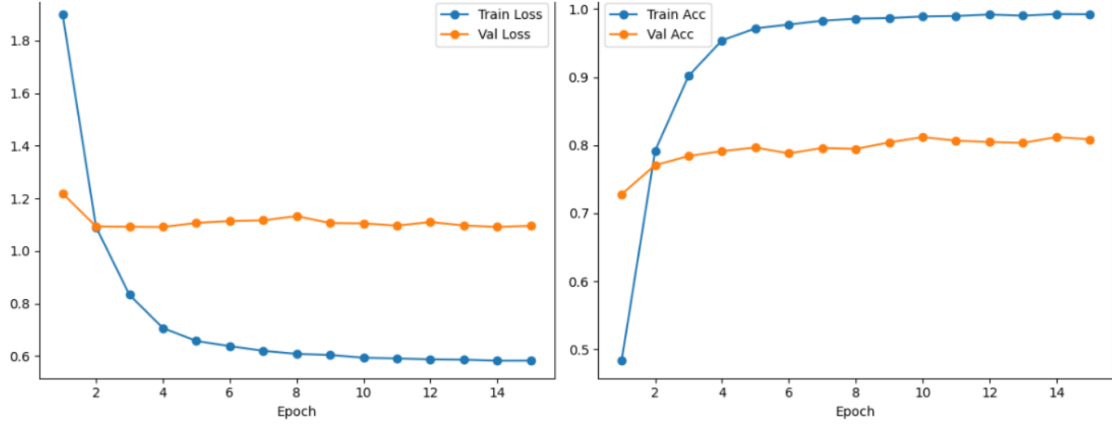


Fig. 3. Training and validation loss (left) and accuracy (right) over 15 epochs.

TABLE V
RETRIEVAL PRECISION@K FOR EMBEDDING-BASED RECOMMENDER

K	Precision@K
1	90%
3	85%
5	80%

accuracy of 80.9% and demonstrated resilience to common distortions, with accuracy dropping to 73.8% under synthetic noise. The extracted 1 280-dimensional embeddings powered an effective top-5 visual recommender, achieving up to 90% Precision@1. Key contributions include systematic evaluation of EfficientNet V2 on in-the-wild RGB clothing images, com-

prehensive noise-robustness testing (blur, brightness, background), a purely visual retrieval module, eliminating the need for metadata, and an end-to-end framework suitable for real-world e-commerce applications. Future work could explore domain adaptation for cross-dataset generalization, real-time optimization for mobile deployment, and rigorous user studies to quantify recommendation satisfaction.

DECLARATIONS

The authors used the generative AI tool ChatGPT (OpenAI, San Francisco, CA, USA) to improve the language and clarity of the manuscript. The authors reviewed and edited all content generated by the tool and take full responsibility for the final version of the manuscript.

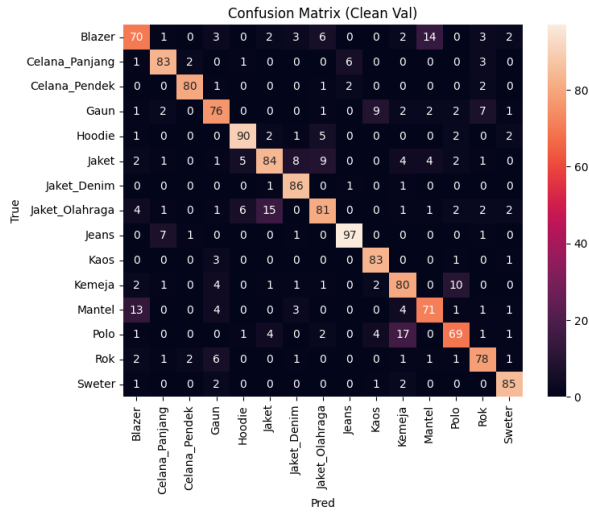


Fig. 4. Confusion matrix visualizing per-class performance on the clean validation set (1 500 images).



Fig. 5. Example clean (right) vs. noisy (left) test image.

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